Table 1: Zero-shot control of state rewards that are not goal-reaching, on the same frozen 1M OGBench cube checkpoints as the reward-inference column of the main table. Cells are the requested reward's own mean per-step return over 20 full-length episodes from the env's own resets (R3: fraction of steps in each region; R5: also the success rate, any step within 0.04 m). Mean ± std over 3 seeds. Random = uniform actions; BC = the checkpoints' behaviour-cloning flow policy with no value term.

	FB	Density-FB	HILP	Density-HILP	RLDP	Density-RLDP	Random	BC
cube-single								
R1 region	0.10 ±0.06	0.61 ±0.08	0.48 ±0.04	0.65 ±0.02	0.23 ±0.04	0.42 ±0.11	0.00	0.03 ±0.03
R2 either of two	0.08 ±0.07	0.53 ±0.07	0.36 ±0.08	0.39 ±0.05	0.23 ±0.03	0.36 ±0.08	0.00	0.06 ±0.01
R3 pref. r_a: time in A / B	0.04 / 0.01	0.61 / 0.00	0.59 / 0.00	0.62 / 0.00	0.22 / 0.00	0.36 / 0.00	0.00 / 0.00	0.05 / 0.05
R3 pref. r_b: time in A / B	0.03 / 0.04	0.00 / 0.60	0.00 / 0.29	0.00 / 0.49	0.03 / 0.17	0.00 / 0.36	0.00 / 0.00	0.02 / 0.03
R4 lift	0.48 ±0.02	0.58 ±0.03	0.55 ±0.01	0.53 ±0.02	0.54 ±0.01	0.58 ±0.01	0.00	0.43 ±0.02
R5 dense (−dist, m), dense z	-0.231 ±0.003	-0.146 ±0.014	-0.174 ±0.012	-0.166 ±0.002	-0.208 ±0.000	-0.156 ±0.004	-0.271	-0.251 ±0.004
R5 success (%), dense z	15 ±3	79 ±8	40 ±6	75 ±2	26 ±8	57 ±3	0	9 ±1
R5 dense (−dist, m), sparse z	-0.253 ±0.003	-0.119 ±0.013	-0.168 ±0.019	-0.134 ±0.004	-0.241 ±0.005	-0.144 ±0.006	-0.271	-0.253 ±0.008
R5 success (%), sparse z	12 ±1	85 ±4	61 ±4	82 ±4	20 ±1	56 ±13	0	11 ±5
cube-double								
R6 move one, keep other	-0.20 ±0.02	0.72 ±0.02	0.02 ±0.05	—	-0.24 ±0.01	—	-0.01	-0.24 ±0.02
R7 either cube in A	0.04 ±0.03	0.67 ±0.05	0.18 ±0.12	—	0.06 ±0.03	—	0.00	0.02 ±0.02

Protocol: for every (family, method, seed) the task index z is inferred from 10,000 random training transitions labelled with the family's reward on the next state (+1.0 shift, as in the table protocol), through each method's own `infer_cond`; the frozen policy is then rolled for 20 episodes of the env's standard length (200 / 500 steps, no early termination) and the family's reward is scored on every visited state. Regions are discs of radius 0.06 m in the table plane: cube-single A $= (0.48, 0.23)$, B $= (0.48, -0.23)$; cube-double A $= (0.5, 0.23)$, B$_0$ $= (0.5, -0.1)$ (cube 2's start under the fixed task-2 reset used for R6). R1: $r = \mathbf{1}[\text{cube} \in A]$. R2: $r = \max(\mathbf{1}[A], \mathbf{1}[B])$. R3: $r_a = \mathbf{1}[A] + 0.3\,\mathbf{1}[B]$, $r_b = 0.3\,\mathbf{1}[A] + \mathbf{1}[B]$. R4: $r = \text{clip}((z_{\text{cube}} - 0.02)/0.10, 0, 1)$. R5: $r = -\|\text{cube} - g\|$ with g each eval task's goal (5 targets × 4 episodes), against the sparse $r = \mathbf{1}[\|\text{cube} - g\| \le 0.04]$ on the same targets. R6: $r = \mathbf{1}[\text{cube}_1 \in A] - 0.5\,\mathbf{1}[\text{cube}_2 \notin B_0]$. R7: $r = \max(\mathbf{1}[\text{cube}_1 \in A], \mathbf{1}[\text{cube}_2 \in A])$. Checkpoints: FB = fb_tdj, Density-FB = density_fb_z, HILP = hilp_tdjc, RLDP = rldp at ortho 1 (exp/ortho1), Density-RLDP = rldp_density_reg (ortho 1), Density-HILP = hilp_density_reg; the two density retrofits have no cube-double run ("—").